Historical research report

Round 1: Independent diligence and process audit

Review date: 4 November 2026 | Original report: 19 pages

Historical snapshot

This file preserves an earlier review. Later findings and the current selection supersede its recommendations, rankings and proposed next actions. Read the current shortlist and Round 4 before using this report for a development decision.

Current assessment

Current shortlist
github.com/KyleJKistner/ddiscov/blob/main/decisions/current-shortlist.md

Round 4 research and decisions
github.com/KyleJKistner/ddiscov/blob/main/research/rounds/round4/README.md

Original report preserved

All original report pages follow unchanged. The original report page numbers are retained; this archival cover is additional.

INDEPENDENT RESEARCH AND PROCESS AUDIT

ddiscov
From promising results to developable assets

Decision report | 4 November 2026 UTC | Repository status reviewed through 3 November 2026

Main finding. Your concern is supported, but the current repository already recognizes many scientific limitations. The unresolved problem is converting those cautions into decisions, access arrangements and executable work. Repeated searches can keep finding plausible experiments without establishing who will let you develop the resulting asset.

Recommended next allocation of attention

Priority for next action	Recommendation
HJ-02 and SYM2081, in parallel	Establish actual owner interest and a proportionate evaluation/option route. HJ-02 is the clearer licensing inquiry; SYM2081 may offer the more economical first falsification experiment.
API5	Ask NYU to identify a useful, nonduplicative development contribution around its current program.
Icerguastat and ODAD1	Treat the former as an existing-company partnership and the latter as a delivery-platform problem. Neither is a confirmed independent startup package.
DS-1211 and new scar leads	Retain bounded diligence questions. Do not promote them merely because they have received less scrutiny.

Funding conclusion. This review has not verified an "obviously fundable" company package. It has identified credible candidates for a bounded diligence milestone. A small investment can legitimately buy proof and access; it need not wait for every development risk to disappear. What it needs is a plausible way to capture the value of a decisive result.

Three parallel diligence tracks examined the presumed leading candidates and then challenged alternatives. This report integrates that work with an audit of the complete repository snapshot and fresh primary-source checks. The accompanying JSON register separates facts, unknown access conditions and proposed decisions.

Review basis: commit c241dc4c6cfa7cd5312a7237dda61a4d2af50c97. No owner outreach, purchases, wet-lab experiments or remote repository edits were performed. References are linked at the end.

01 / SCOPE AND CONFIDENCE

What was actually audited

The stated program seeks restoration of function using existing drugs, proteins and genetic therapies. Its first milestone is a reproducible preclinical result that a development partner can use. That is the profile applied here; no undisclosed budget ceiling, preferred modality or exclusive orphan-disease mandate was invented. [R1]

Observed in this snapshot	What it establishes
28 files at the pinned commit	Complete review of the released edition, not the private underlying research archive.
35 program families; 31 research phases	Substantial breadth, with unequal depths of review.
232 historical topics; 825 occurrences	Coverage and provenance units, not 825 independent investable assets.
70 component rows / 68 concepts; 67 comparison rows / 57 concepts	The project already tracks important duplicates and construct distinctions.
139 of 232 topics have direct links in the compact topic table	Blank compact-index fields are disclosed and may have occurrence-level provenance. They are not proof of unsupported research.

The A-tier qualification

The public edition contains no A-tier ranking. It describes HJ-02, icerguastat and ODAD1 as the three most recently emphasized leads from successive rounds, and leaves portfolio-wide investment order open. Those are the three audited as the presumed A tier. An earlier or private ranking cannot be reconstructed from this edition alone. [R2-R4]

What the code and provenance do support

I reran the included WNT analysis and reproduced the reported RMSEs: 3.4022 versus 1.0528; sensitivity 3.3329 versus 0.9532; second panel 0.4583 versus 0.1525. The repository discloses that the context feature was chosen with qualitative outcomes known and that only four or five assemblies inform each panel. This is a useful reconstruction, not prospective validation of a discovery engine. [R5]

I reviewed the airway analysis code but did not independently rerun its absent source workbook. Its donor-held-out design is appropriate for untreated disease/control separation; that task is different from predicting treatment response. The 116 archive fingerprints identify versions, but their underlying files were not supplied, so I could not replay the original agent searches or all archived decisions. [R5-R6]

02 / PROCESS DIAGNOSIS

Why the search can keep circling

1. The output unit is often an experiment, while the desired outcome is an opportunity

A study worth doing can still be commercially inaccessible, already assigned to another team, or irrelevant to the next financing decision. Every shortlisted record should say which development route it supports: license, option-backed project, co-development, paid research, or acquisition. The same biology can support one route and fail another.

2. Uncertainty is documented more consistently than it is closed

The decisions page already separates biology, feasibility, contribution and transfer. It correctly flags the sham comparator for HJ-02, the existing icerguastat program and the mature-airway challenge for ODAD1. The improvement needed is an operating rule: unresolved conditions must determine the next action and the spending limit. Adding another caution paragraph should not leave promotion unchanged. [R2]

3. Unequal scrutiny can make the newest idea look best

The repository explicitly notes successive search rounds and differing amounts of review. That creates a plausible selection bias: old candidates accumulate known weaknesses while new ones retain undiscovered weaknesses. Apply a common challenge packet to all finalists before comparing them. Revisit an older lead when its next uncertainty is cheaper or more decisive to resolve. [R2]

4. Evidence can drift across identity, delivery and timing

A related peptide is not HJ-02; basal-cell transduction before differentiation is not mature-airway delivery; drug in culture medium is not topical skin delivery. The repository often preserves these distinctions well. Make them mandatory fields in the registry so later summaries cannot silently combine separate claims. [R2,R7]

5. A feasible measurement can conceal model-development work

A catalogued cell line and an available sequencing service do not establish a qualified disease model. A morphology signal may fail to predict transport, force, conduction or durable tissue function. Model qualification should be a paid, explicitly fallible milestone before efficacy claims are scored.

6. A service subtotal can become an unrealistic project expectation

The repository accurately calls its icerguastat sequencing and specimen-preparation prices a two-service subtotal. The model, operator, exposure analytics, electrophysiology, failed batches and access terms still require pricing. Keep these exclusions visible in every decision view. [R2]

7. More public searching has diminishing value at a private-information boundary

No search engine can verify confidential license terms, present inventory, a private negative study or what an owner would pay for. Once those are the decisive uncertainties, the next productive step is a defined owner inquiry or experiment. A new candidate should not be a device for postponing that step.

03 / ACCESS AND VALUE CAPTURE

Replace one availability flag with seven facts

Question	Evidence needed for the next stage
Exact material	Structure or sequence, stereochemistry, construct, formulation, reference lot and activity specification.
Research supply	A confirmed source, quantity, lead time and quality documentation for the proposed experiment.
Permitted use	Actual terms covering this entity, this operator and this experiment, including contractor use where relevant.
Commercial route	Current owner/licensee and a plausible field, territory or role. Early evaluation terms can precede a full license.
Human models	Specific donor/genotype, inventory, independent biological units and permission for the intended use.
Result ownership and transfer	Rights to raw data, derivatives, publication and transfer to a future partner; obligations to the material supplier.
Execution	An operator willing and able to perform a qualified assay, with a complete scope and milestone quote.

Record each field as documented, owner-confirmed, contractually secured, or unknown. Keep the source date separate from the date it was checked. "No answer yet" is a state of knowledge, not a conclusion that an asset is unavailable.

Avoid symmetric access mistakes

A licensing listing is an invitation to discuss a package, not proof that exclusive rights in the desired field remain available. Conversely, an existing licensee can be the correct partner rather than a reason to reject the biology. NIH's published workflow explicitly separates inquiry, application, negotiation and executed agreement. [S37]

Research-use restrictions must be read for the actual material and transaction. Addgene offers some materials to industry under specific terms; that does not imply every academic plasmid is available for commercial development or that every research-only label forbids all company research. [S38-S39]

Patent search is also evidence-specific. A named gene in a disclosure does not establish a blocking claim. A PCT status label does not resolve all national rights. Examine claims, family members and assignments with the proposed product in view. This report does not supply a freedom-to-operate opinion. [S40]

Proportionate threshold

Before modest feasibility spending, require a credible route to value capture and usable research/data terms. Before a substantial study, require a stronger owner-aligned scope and commercial path. Requiring a completed license, complete target identification and all efficacy evidence before any seed financing would make the process unnecessarily restrictive.

04 / PRESUMED LEADING CANDIDATE 1

HJ-02: first concrete licensing inquiry

Decision: retain as a conditional option-and-replication project. It is the clearest first owner discussion among the presumed three, not a demonstrated best-return company.

Evidence that matters

The cardiac study dosed after infarction and reports early scar and provoked-rhythm signals. The day-15 pump-function panels used sham as the Dunnett reference, with five animals per treatment group and four sham animals. A nonsignificant difference from healthy sham does not establish superiority to injured vehicle or equivalence to health. [S3]

The published chemistry work does not establish direct PTPsigma catalytic inhibition or a confirmed binding target. Original dissertation work also describes unresolved target-identification efforts. A phenotypic program can still be fundable; these facts change the exposure, reproducibility and safety questions. [S4-S5]

The later 40-day experiment used ISP, a different intervention. It cannot establish HJ-02 durability, and its negative or convergent findings should not be assigned directly to HJ-02 either. [S6]

A positive access update

OHSU explicitly offers licensing or co-development. Its current page lists US12,729,177; the USPTO grant index confirms issuance on September 8, 2026. This improves the specificity of the access inquiry. It does not establish available field, price, inventor support or inventory. [S1-S2]

First milestone: define the package before rebuilding it

Ask for updated animal-level data, all follow-up and negative studies, current chemical series, exposure information, safety observations, available material and the complete patent schedule. Establish whether independent replication is a result OHSU would actually use and which rights or option could accompany that work.

Second milestone: one interpretable durability experiment

Use authenticated material and reproduce the proximal functional assay before a major animal study. The main experiment should randomize or stratify after measuring initial injury, blind assessment, compare with injured vehicle, and follow meaningful cardiac function after dosing stops. Rhythm assessment should address the intended clinical claim. HJ-01 can serve a useful chemistry comparator if its inclusion changes the interpretation.

The protocol needs an effect threshold and uncertainty bound chosen before results, not a favorable comparison selected afterward. A prespecified useful effect against injured vehicle is the key investment readout. Scar staining alone or resemblance to sham should not substitute for it.

What stops the project

Hold substantial spending if there is no credible commercial route, no reliable active material, an exposure/safety problem that prevents a meaningful test, or no identifiable development or financing decision the result would change. Stop this development thesis if an appropriately designed replication fails its useful functional threshold. Do not demand target elucidation as a universal prerequisite.

05 / PRESUMED LEADING CANDIDATE 2

Icerguastat: a partnership before a new assay

Decision: pursue only a defined licensee-led transaction or development role before committing to a costly independent human model.

The commercial fact is already concrete

UChicago announced an exclusive Enemsa license to the US10,905,663 family for demyelinating diseases including CMT in July 2026. Enemsa publicly seeks partners. The useful question is what role, field or data contribution it will offer, not whether the molecule is unclaimed. [S7-S8]

Biology deserves credit without inflation

The R98C mouse study used post-onset treatment, blinded evaluations and meaningful functional cohorts. It supports partial benefit. Using the repository's approximate means, the conduction change from 14.5 to 16.4 m/s is about 13% above diseased baseline and 11% of the gap to healthy mice at about 32 m/s. These are calculations from approximate plotted means, not raw-data reanalysis. [R2,S9]

The sponsor's exploratory ALS report distinguishes an intent-to-treat trend from a significant per-protocol slope comparison. Human exposure experience is useful; it is not demonstrated CMT efficacy. Obtain the full clinical report rather than inheriting a promotional efficacy label. [S10]

The proposed human study has a qualification phase

The NYSCF list identifies R98C and corrected-control lines, but catalogue entries do not establish current supply, commercial terms or independent donors. The early nerve-on-chip demonstration is not itself validation of an MPZ disease assay. [S11-S12]

Before testing drug efficacy, an operator should show a reproducible genotype-associated functional deficit, adequate myelination, assay stability and sensitivity to a meaningful improvement. An isogenic control strengthens causal interpretation, while independent donors address generalization. Repeated differentiations and sequencing samples must remain nested within their donor backgrounds.

Measure the relevant free exposure in the assay and relate it to human pharmacology. Distinguish an increase in propagation from a change in myelin. Functional gain should be the primary development readout; one stress-response marker is insufficient to resolve the published mechanistic dispute. [S13]

The cost correction is about scope

The repo's academic $9,000-10,200 and industry $11,940-14,580 figures cover sequencing and specimen preparation only. They are correctly labeled. No full project quote was verified here. Require separate pricing for access, differentiation, model qualification, electrophysiology, analytics, labor and failed batches. [R2]

The decisive gate

Enemsa should identify a decision that this study would change and a useful economic or development role for its funder. If it cannot, a technically elegant assay may create value for an existing program without creating an opportunity for you. A paid or rights-bearing collaboration remains a legitimate result; founding a separate company need not be the only acceptable route.

06 / PRESUMED LEADING CANDIDATE 3

ODAD1: a delivery problem remains central

Decision: retain as a defined research or delivery-platform collaboration. The public package does not yet justify an independent, near-term development asset.

What the biological result establishes

The organoid study introduced full-length ODAD1 before differentiation and restored substantial beating in two non-founder patient genotypes. The recorded result concerns local ciliary motion. It does not demonstrate delivery to mature founder-variant airway tissue or restoration of coordinated mucus transport. The repository already makes this distinction. [R2,S14]

Why the distinction changes the investment

The intended product must reach the relevant cells in an established airway, express an appropriate construct, preserve tissue health and improve a directional transport function. Genetic complementation under a permissive culture protocol is encouraging evidence for the payload; it is not evidence that the delivery system has solved those tasks.

A proposal to add mutant suppression should follow a fair replacement-only test at measured coverage and expression. Otherwise, failure of delivery or cell maturation could be misread as a requirement for a more complex therapeutic architecture. Compare genotypes without assuming that evidence from one variant transfers to another.

Availability has several independent custodians

The repository identifies banked material for the comparison genotypes, with one documented donor per genotype and an independent founder donor unresolved. Tissue access, clinically useful delivery, vector/material terms and transferable data still need to be combined into one executable package. An earlier inhaled-RNA disclosure names ODAD1, requiring claim-level investigation rather than automatic rejection. [R2,S15-S16]

A useful next experiment, if the platform is real

First qualify mature tissue and a delivery method that can generate interpretable coverage. Then compare replacement and controls with measured expression, viability and epithelial integrity. Use directional particle or mucus transport as the functional outcome, with beat frequency and waveform as supporting measurements. The biological unit is the independent donor, not each organoid, field, video or particle.

The airway code in the repository offers a helpful donor-level analytical precedent. Its disease/control classification does not establish that a treatment will restore transport. A treatment experiment needs paired or randomized genotype-matched tissue and appropriate analysis of within-donor replicates. [R5]

What would change the recommendation

UCL lists a base-editing project targeting the recurrent CCDC114/ODAD1 c.742G>A allele and says affected-patient cultures are already characterized. This is a competing research direction and a potential partner, not evidence of successful editing or exclusive rights. [S41]

Search both gene names and transcript forms: legacy c.742G>A corresponds to MANE NM_001364171.2:c.853G>A. An alias table can prevent missing relevant science, material custodians and programs. [S42]

07 / ALTERNATIVE EXAMINED FURTHER

SYM2081: revisit with a precise product test

Decision: run access diligence beside HJ-02. This may be the best candidate for a relatively economical first falsification study, subject to an actual operator scope and quote.

Keep the published experiments separate

Human skin received drug through culture medium, not topical epidermal dosing. The DNFB topical experiment treated before sensitization. The rosacea-like experiment co-injected drug and LL-37; it was not topical treatment. Fc-epsilon-receptor-mediated degranulation was unaffected. The paper reports 100 micromolar topical formulation, while the patent example says 5.0%; the original records must reconcile these descriptions. [S18-S19]

Original receptor pharmacology supports activity at GluK1 as well as GluK2. A reproducible phenotype can still justify development, but a unique GluK2 selectivity claim should not be inherited without qualification. [S20]

Why it remains interesting

Pitt identifies a specific technology and an institutional contact route. Published topical-use claims make a license discussion more relevant than assuming a new formulation creates unconstrained commercial space. Neither a listing nor a research reagent establishes exclusive availability. [S17,S19]

The unresolved delivery question can be tested directly within a human-tissue experiment. Compared with a long cardiac remodeling study or a new MPZ nerve platform, that may have a favorable information-to-cost ratio. This is an inference about scope, not a price quote or a demonstrated cost advantage.

Proposed first study

Use independently identified donors and paired skin conditions. Compare topical formulation, dermal-bath positive control and matched vehicles. Measure tissue exposure, quantitative mast-cell response, viability and barrier integrity together. Authenticate stereochemistry, formulation and dose first. Blind the analysis and retain donor-level results rather than pooling fields as independent samples.

Result pattern	Decision it informs
Bath activity; no topical activity	A delivery/formulation limitation. Do not conclude the biological hypothesis is disproved.
No reproducible bath activity	Recheck material and assay, then reconsider the biological premise before formulation expansion.
Topical activity with preserved tissue integrity	Proceed to a disease-relevant comparison and a defined product-development plan.
Apparent suppression with tissue damage	Do not classify the readout as useful therapeutic activity.

Select a specific indication and align treatment timing with the claim. A prevention product may be valid if it has a useful clinical setting.

07A / SYM2081 FOLLOW-UP

Recent clinical failures raise the differentiation bar

The alternative review included current pathway competition, not only the original academic result. It uncovered adverse clinical evidence that should remain beside the SYM2081 proposal in every funding discussion.

Program and source	Observed result	Scope of inference
Oral EVO756, CSU [S43]	The 160-participant phase 2b trial missed its primary endpoint; the sponsor did not support further CSU development.	Direct negative evidence in CSU for this MRGPRX2 antagonist.
Oral EVO756, atopic dermatitis [S44]	The 121-participant phase 2b trial missed primary and secondary endpoints; AD development was stopped in September 2026.	Negative dermatology translation; the molecule continued in migraine.
Oral EP262 / INCB000262, CSU [S45]	The October 2026 report found no significant efficacy at either tested dose in 113 patients; a planned lower-dose part was terminated.	Another controlled negative result for direct MRGPRX2 antagonism.

What this changes

These studies did not test topical SYM2081, GluK modulation or a rosacea-specific product. They therefore do not falsify that proposal. They do reduce confidence in a broad argument that dampening an induced mast-cell readout will necessarily improve a dermatologic disease.

A convincing program now needs to explain what differs in its clinical population, local exposure, mechanism or functional effect. Those differences should produce testable predictions. A pathway label or a claim that the topical route is inherently better is insufficient.

Effect on the first study

Retain the paired human-skin delivery/phenotype study as an economical way to falsify a key premise. Where the assay supports a fair comparison, include a direct MRGPRX2 antagonist. If both interventions equally suppress the experimental challenge, that alone will not distinguish SYM2081 from the clinical failures. A later disease-relevant endpoint and patient-selection rationale are needed before a larger financing claim.

This lowers confidence in SYM2081 as a company opportunity while leaving the value of a small, well-designed falsification experiment intact. The ranking remains a choice of next actions, and HJ-02 may be the better package if its owner data and terms are more favorable.

08 / ALTERNATIVE EXAMINED FURTHER

API5: begin with the owner's current program

Decision: third targeted inquiry. The immediate opportunity is to agree on a useful development contribution, not automatically to repeat the published parent-protein work.

A relevant human starting point

The Nature study includes human risk-genotype organoid protection and Paneth-cell recovery, alongside mouse work. This is meaningful human-material evidence, while the intended functional, timing and delivery claims still require separate tests. [S22]

The live development page changes the assignment

NYU's CAD01-05 listing describes a synthetic peptide approach and active internal accelerator work on other IBD subtypes, minimal functional fragments and basic pharmacokinetics/toxicology. It also reports applications covering fragments and uses. A generic proposal to optimize API5 or test a new disease model may overlap work already assigned. [S21]

Ask for the decision, then choose the experiment

The current lead fragment is a sensible starting inquiry, but should not become an arbitrary prerequisite. NYU might value independent parent-versus-fragment benchmarking, delivery work, a new model, a comparative study or a differentiated analogue. Any of these can be worthwhile if tied to accessible material, a defined development question and usable rights to the outcome.

Obtain the exact construct, potency assay, manufacturing state and existing data under suitable terms. Determine what remains uncertain after the private package is considered. Ask which result would change licensing interest, candidate selection or readiness for the next stage, and who would use that result.

A conditional study concept

For an established-disease repair claim, compare the intended therapeutic format with the parent benchmark after injury is present in independently derived human tissue. Use functional epithelial integrity or restitution alongside survival and cell-type measurements. Relate activity to achievable exposure and duration. Preserve genotype, donor, batch and assay-condition distinctions.

A strong survival result could support a protective mechanism while leaving the clinically useful repair claim unanswered. Conversely, a properly chosen local functional readout could establish value before an exhaustive receptor-level mechanism is known. The experiment should answer the decision actually facing the owner rather than maximize the number of attractive markers.

Proceed or hold

Proceed if an attainable development arrangement, accessible material and a nonduplicative milestone emerge. Hold if the proposed experiment is already completed, the required construct cannot be shared, the data cannot be transferred, or success would still not change a development decision. The owner's safety and differentiation language is a development hypothesis, not established clinical superiority.

09 / ALTERNATIVES WITHOUT AUTOMATIC PROMOTION

Further candidates and a material clinical update

DS-1211: review the existing clinical asset

The 65-participant phase 2 record emphasizes safety, pharmacodynamics and exposure. It does not demonstrate improved patient function. Daiichi disclosed discontinuation in October 2025, citing timeline and competitiveness. A discontinued program is not necessarily an available transaction. [S31-S32]

A new fact matters to the broader premise: BioMarin's ENERGY 3 study increased circulating PPi without improving its rickets co-primary endpoint; the company discontinued BMN 401 in July 2026. Different disease, modality and endpoints prevent direct extrapolation to DS-1211/PXE. The result nevertheless weakens any assumption that PPi correction alone establishes clinical benefit. [S33-S34]

The next valuable work is owner-authorized review of individual human PK/PD, safety, all exploratory outcomes and the original termination decision. Only then assess a differentiated trial and actual transaction terms. A generic additional mouse study would have low value if the financing bottleneck is still the clinical path.

SLIT2 and CBP/EP300: split the products

Stanford's S23-520 provides a live institutional lead and mouse scar-prevention biology. The inspected human evidence concerns tissue expression rather than treatment efficacy. Keep the protein and chemical-probe branches separate. [S23-S25]

The patent's aa26-900 reagent description appears to match a mouse SLIT2 product; the listed human reagent has a different fragment length. Confirm that inference with the investigators and bridge the intended human product. I-CBP112 is a CBP/EP300 bromodomain probe, not a validated selective EP300 clinical asset. [S25-S28]

Verteporfin is a substantive comparator: primary work includes pig wounds, and the official CTIS record lists an authorized Phase 2 scar-prevention study. Do not infer trial enrollment or clinical efficacy from authorization. [S29-S30]

An accessible product with a useful treatment window, wound quality and manufacturing advantage could still be valuable. Novelty of pathway alone does not establish that advantage. A fresh, lightly reviewed mouse lead should not displace an older candidate simply because its limitations are less familiar.

GAS6/AXL: early screen only

A Stanford listing and primary mouse repair study provide another route to investigate. Human observational relevance is not treated-human efficacy. No full product, rights or execution package was verified in this screen, so this was not promoted to the leading group. [S35-S36]

These comparisons illustrate a more productive search rule: maintain a reserve of identifiable opportunities, but spend deep diligence effort only where the next evidence could change a concrete development decision.

10 / RECOMMENDED OPERATING PLAN

Choose the next commitment

Step	Concrete deliverable	Decision
1. Owner package checks	Parallel OHSU and Pitt inquiries; NYU next. Ask for current scope, private updates, materials and an evaluation path.	Continue only where a useful role is plausible.
2. Common evidence review	Exact product, strongest functional result, strongest adverse result, independent unit and relevant comparator.	Compare candidates at equal scrutiny.
3. Operator feasibility	Written assay capability, qualification milestone, material requirements, complete quote and failure assumptions.	Choose the first experimental tranche.
4. Proportionate agreement	Material use, data ownership/transfer and credible commercial path. An option may remain conditional.	Protect the value of the proposed result.
5. Decisive experiment	Predefined useful effect, uncertainty, controls and stop/go rule.	Advance, redesign once for a specific reason, or stop.
6. Financing package	Actual rights position, functional evidence, development differentiation, next budget and accountable team.	Make a company or collaboration proposition reviewable.

What should be compared when the responses arrive

HJ-02 may win on asset definition and licensing maturity. SYM2081 may win on the cost and diagnostic clarity of its first experiment. API5 may win if NYU offers a mature current fragment and an open milestone. These are conditional comparisons. Owner responses and actual quotes should be able to change the order without rewriting the scientific history.

Keep funding routes explicit

A sponsored study, an option-backed development project and a standalone venture have different return mechanisms. A strong partner may remove technical risk while limiting your economics; a broad license may add ownership while leaving more development cost with you. Compare the package you can obtain, not the most attractive hypothetical version of each candidate.

When to stop expanding the search

Pause broad intake while the leading uncertainties are owner intent, material terms or assay feasibility. Resume it if the realistic packages fail the desired economics or scientific threshold. A hold is reversible when new evidence arrives; it should not require rediscovering the whole field.

11 / MAKE THE FIRST EXPENDITURE REVIEWABLE

Two conditional statements of work

HJ-02: package review followed by functional replication

Inputs: exact active compound, authenticated reference, chemical and exposure information, raw historical outcomes, prespecified intended product claim, and a usable research/commercial route. Request all negative or null studies, not only the flagship paper.

Qualification: reproduce the agreed proximal functional assay and verify exposure and material consistency. If this fails, diagnose material or assay failure before treating it as a clean test of the therapeutic hypothesis.

Decision study: injury-balanced, blinded, randomized comparison against injured vehicle, with a useful pump-function endpoint and post-withdrawal follow-up. Choose rhythm methods that fit the intended claim. Agree how deaths, exclusions and repeated observations are handled before unblinding.

Deliverable: individual-level data, complete protocol and deviations, compound records, analysis code, effect estimate with uncertainty, and a recommendation against a prespecified minimum useful effect. Do not specify a numerical sample size without variance and effect assumptions. Budget the experiment that can change the decision, not merely a pilot that will predictably require repetition.

SYM2081: delivery and phenotype in the same human tissue

Inputs: reconciled formulation record, exact stereoisomer and activity specification, tissue rights and independent donor identities. Define whether the initial product is intended for prevention or established inflammation.

Qualification: establish a reproducible bath-response benchmark, skin integrity and analytical recovery. A failed positive-control system cannot judge topical delivery.

Decision study: paired donor comparison of topical candidate, bath benchmark and matched vehicles, with measured dermal exposure and quantitative functional response. Include tissue viability and barrier integrity so suppression through damage cannot count as success. Add the disease-relevant comparator when the indication is selected.

Deliverable: donor-level response and exposure data, formulation preparation records, stability observations, blinded scoring, variance estimate and a practical interpretation of the four outcome patterns in the SYM2081 section. An initial feasibility study can inform a subsequent powered disease-model test; it should not be advertised as clinical proof.

Quote requirements for both

Price materials, model setup, labor, assay qualification, analytical methods, failed runs, data delivery and contingencies separately. Specify what constitutes a valid run and what is payable after failure. Include licensing and other external costs where applicable. No full quote was obtained in this review, so no invented total budget is presented.

12 / RESEARCH WORKFLOW AND REPOSITORY CHANGES

Use agents to resolve decisions

The main value of parallel research is independent challenge, not multiplication of candidate lists. Assign every finalist the same work packet and require a documented decision consequence for each finding.

Track	Required output
Science reviewer	Exact tested intervention; primary functional result; strongest adverse evidence; treatment stage; experimental unit; exposure and comparator limits.
Access reviewer	Current owner/licensee; attainable development route; material and model terms; source dates; explicit remaining owner-only questions.
Execution reviewer	Operator capability; assay qualification; full scope and quote requirements; study that could resolve the key uncertainty.
Integrator / challenger	Resolve conflicts, separate inference from fact, compare at equal depth, and choose the next action rather than average away fatal unknowns.

Small repository changes with high practical value

1. Add a current candidate register. The accompanying JSON is a ready starting point, not a replacement for the narrative atlas. It has nine separate product records, sources, action classifications and explicit unconfirmed fields. SLIT2 protein and the CBP/EP300 branch are separate records.

2. Attach a decision consequence to each new finding. Record whether it changes the product claim, owner route, assay, budget or disposition. Keep a dated history so a previously rejected candidate cannot return under a slightly different name without explaining what changed.

3. Use a precise identity key and aliases. Molecule/sequence, formulation, route, indication/genotype, treatment timing and endpoint should travel together. Search old gene names, compound codes and transcript-specific variants. The ODAD1/CCDC114 example shows a practical recall failure this can prevent. [S41-S42]

4. Record source and verification dates separately. A recently crawled old technology page is not a recent ownership confirmation. Sponsor announcements, registry records and published studies answer different questions.

5. Correct one visible wording issue. The SYM2081 atlas description should specify drug added through culture medium rather than imply an epidermal topical experiment. The follow-up evidence record should preserve the distinct routes used in the dermatitis and rosacea experiments. [R7,S18]

6. Keep code claims proportional. Preserve donor/scaffold/construct grouping and prospective validation limits. Retrospective classification, mechanism reconstruction and experimental candidate selection are different tasks. The current methods documentation already does much of this correctly. [R5]

The provided register deliberately avoids invented success probabilities and composite investment scores. A scientific score cannot compensate for an absent right to develop the asset; an attractive licensing page cannot compensate for a nonfunctional experiment.

13 / PREPARED, NOT SENT

Owner inquiry briefs ready for use

OHSU: technology 2944 / HJ-02

Purpose: determine whether an independent functional replication program can support an attainable development opportunity.

Ask which cardiac-repair rights or co-development structures are currently open; whether any existing options or licenses restrict the proposed scope; and how US12,729,177 fits the current family. Request the current lead series, authenticated material route, full animal-level data and subsequent efficacy, exposure and safety studies. Ask specifically whether a blinded durability study against injured vehicle would change the owner's next decision, and whether an evaluation license or option could cover that work.

Useful response: a named business and scientific counterpart, a current data-room index, material terms, a proposed development role and an evaluation framework. A general expression of enthusiasm is not enough to price the next experiment.

Pitt: technology 06272 / SYM2081

Purpose: determine whether a topical human-skin delivery/function study is both feasible and commercially useful.

Ask which rights, territories and indications remain open and whether a license or option could cover the intended formulation. Request the exact stereoisomer, reference material, formulation and application records; reconcile the patent's 5.0% description with the paper's 100 micromolar formulation. Ask for donor identifiers and pairing information, unpublished topical human-skin or established-disease studies, and any current work that would duplicate the proposed experiment.

Propose a paired human-skin comparison of topical delivery and bath benchmark, with exposure, mast-cell response and tissue-integrity measurements. Ask which initial indication and comparator the investigators regard as credible, and what outcome would change their licensing or candidate-selection decision.

Useful response: defined material and formulation, coherent dosing record, tissue/operator route, open milestone and attainable terms.

Why these are questions, not a pre-negotiated project

A private package may already resolve a public uncertainty or reveal a decisive failure. Ask for that package before locking in a new experiment. The early conversation should be sufficiently specific to obtain a meaningful response while leaving room for the owner to propose a better development contribution.

No messages have been sent. These briefs are prepared for explicit authorization of external outreach. Public sources establish identifiable counterparties; they do not establish the availability of the requested arrangement.

14 / MOVE FROM FACTS TO COMMITMENTS

Additional inquiry briefs and the financing test

NYU: CAD01-05 / API5

Ask which construct or comparative program is current, what the internal accelerator has completed, and which milestone remains open to an outside developer. Request identity and potency information, access to parent and current fragments where appropriate, existing human-donor and animal data, and the status of PK/toxicology work. Ask whether a post-injury functional comparison, a delivery study or another defined contribution would be useful. Establish material, data-transfer and commercial terms before proposing a broad independent optimization program.

Enemsa: icerguastat in MPZ-associated neuropathy

Ask what CMT collaboration, sublicense or paid research role the company would consider and which patient-selection or clinical decision an outside human functional model could change. Request the existing genotype-specific package and any similar completed/planned assays. Determine whether the company would provide drug, human exposure data and necessary upstream rights. If it wants assay development, agree qualification criteria and economics before generating efficacy claims.

The financing test

A defensible early proposition should answer five questions in a compact package: What exactly is the product? Why is its functional signal credible? How can the team obtain a useful position in it? Which affordable result changes the next decision? Who can deliver that result, and who will value it? These questions can be answered progressively; they need not all be contractually final before exploratory work.

Reason about the next action in terms of the probability that it changes a decision, the value of that change, and its cost and delay. Do not assign numerical probabilities merely to make a spreadsheet look quantitative. The present record supports ranking information needs more strongly than ranking company valuations.

What is complete, and what remains open

Complete: review of the released repository, targeted rechecking of the presumed three leads, additional alternative-candidate diligence, a reproducible methods check, a new candidate/action register and concrete study/inquiry briefs.

Open: owner willingness, private data, usable material and data terms, qualified operators, complete quotes, actual experimental results and investor appetite. These are not details that another general literature scan can verify. The strongest immediate recommendation is to open OHSU and Pitt discussions in parallel and compare the actual packages, with NYU as a third route.

Confidence: high that the public edition does not establish a confirmed developable package for the presumed A tier; moderate that HJ-02 and SYM2081 deserve the first action slots; low in any ranking of eventual therapeutic success or investment returns before owner data and experiment results. Lack of a verified package is not proof that no fundable package exists.

REFERENCES / 1

Sources and verification basis

Primary studies, official owner/sponsor notices and institutional records are linked below. Repository references are pinned to the audited commit. Public-source verification was performed during this review on 4 November 2026 UTC; publication dates are retained where material. No confidential licensing terms or individual-level unpublished data were obtained.

R1 | Repository scope and stated milestone
Open source github.com

R2 | Development comparisons and study subtotal
Open source github.com

R3 | Discovery crosswalk and denominators
Open source github.com

R4 | Research phases and search history
Open source github.com

R5 | Methods, units and retrospective limits
Open source github.com

R6 | Archive provenance and verification requirements
Open source github.com

R7 | Barrier programs: API5 and SYM2081
Open source github.com

S1 | OHSU technology 2944: HJ compounds and licensing
Open source apps.ohsu.edu

S2 | USPTO September 8, 2026 grant index: US12729177
Open source patentsgazette.uspto.gov

S3 | Sepe et al.: HJ-02 cardiac study, JACC Basic Transl Sci 2022
Open source pubmed.ncbi.nlm.nih.gov

S4 | Blake et al.: acrylamide compounds and pharmacology, 2022
Open source pmc.ncbi.nlm.nih.gov

S5 | Blake dissertation: compound target-identification work, 2022
Open source digitalcollections.ohsu.edu

S6 | Forty-day cardiac follow-up: ISP, not HJ-02
Open source doi.org

S7 | UChicago exclusive Enemsa license announcement, July 30, 2026
Open source polsky.uchicago.edu

S8 | Enemsa partnering: CMT and ALS
Open source enemsapharma.com

S9 | Bai et al.: IFB-088 in CMT mouse models, 2022
Open source doi.org

S10 | Sponsor preliminary ALS results, February 6, 2025
Open source inflectisbioscience.reportablenews.com

S11 | NYSCF/CMTA available iPSC list, March 2026
Open source www.nyscf.org

REFERENCES / 2

Sources and verification basis

Primary studies, official owner/sponsor notices and institutional records are linked below. Repository references are pinned to the audited commit. Public-source verification was performed during this review on 4 November 2026 UTC; publication dates are retained where material. No confidential licensing terms or individual-level unpublished data were obtained.

S12 | Human nerve-on-chip proof of concept, 2019
Open source pmc.ncbi.nlm.nih.gov

S13 | Sephin1 mechanism challenge, eLife 2017
Open source elifesciences.org

S14 | ODAD1 organoid replacement, Cell Discovery 2026
Open source pmc.ncbi.nlm.nih.gov

S15 | PCD donor material and genotype bank, Nature 2023
Open source doi.org

S16 | Earlier inhaled-RNA disclosure including ODAD1
Open source patents.google.com

S17 | Pitt technology 06272: glutamate receptor agonists
Open source inventions.pitt.edu

S18 | Zhang et al.: SYM2081 skin experiments, Sci Transl Med 2024
Open source pmc.ncbi.nlm.nih.gov

S19 | Pitt published application: US20240342136A1
Open source patents.google.com

S20 | Original SYM2081 kainate-receptor pharmacology, 1998
Open source pubmed.ncbi.nlm.nih.gov

S21 | NYU CAD01-05: API5 therapeutic development
Open source license.tov.med.nyu.edu

S22 | Matsuzawa-Ishimoto et al.: API5, Nature 2022
Open source pmc.ncbi.nlm.nih.gov

S23 | Stanford S23-520: SLIT2/ROBO2/EID1/EP300
Open source techfinder.stanford.edu

S24 | Chen et al.: anatomically variant scarring, Cell 2026
Open source pubmed.ncbi.nlm.nih.gov

S25 | SLIT2/EP300 published application WO2026015476A1
Open source patents.google.com

S26 | Mouse SLIT2 aa26-900 reagent 5444-SL
Open source www.bio-techne.com

S27 | Human SLIT2 aa26-1118 reagent 8616-SL
Open source resources.rndsystems.com

S28 | SGC I-CBP112: CBP/EP300 bromodomain probe
Open source www.thesgc.org

S29 | EMA CTIS: SCARFREE-001 verteporfin Phase 2
Open source euclinicaltrials.eu

REFERENCES / 3

Sources and verification basis

Primary studies, official owner/sponsor notices and institutional records are linked below. Repository references are pinned to the audited commit. Public-source verification was performed during this review on 4 November 2026 UTC; publication dates are retained where material. No confidential licensing terms or individual-level unpublished data were obtained.

S30 | Verteporfin porcine and human-xenograft study, 2025
Open source pubmed.ncbi.nlm.nih.gov

S31 | DS-1211 phase 2 official EU results
Open source www.clinicaltrialsregister.eu

S32 | Daiichi FY2025 Q2 reference data: discontinuation, p59
Open source www.daiichisankyo.com

S33 | BioMarin ENERGY 3 primary results, May 18, 2026
Open source www.biomarin.com

S34 | BioMarin BMN 401 program discontinuation, July 30, 2026
Open source www.biomarin.com

S35 | Stanford S23-199: GAS6/AXL repair opportunity
Open source techfinder.stanford.edu

S36 | GAS6/AXL wound study, Sci Transl Med 2025
Open source pubmed.ncbi.nlm.nih.gov

S37 | NIH licensing process: inquiry through agreement
Open source www.techtransfer.nih.gov

S38 | Addgene industry material access
Open source www.addgene.org

S39 | Addgene industry material agreement
Open source www.addgene.org

S40 | USPTO: patent claim scope and definiteness
Open source www.uspto.gov

S41 | UCL founder-specific ODAD1/CCDC114 editing project
Open source www.ucl.ac.uk

S42 | ClinVar: transcript aliases for founder variant VCV000039637.43
Open source www.ncbi.nlm.nih.gov

S43 | Evommune EVO756 CSU phase 2b failure, June 29, 2026
Open source ir.evommune.com

S44 | Evommune SEC disclosure: EVO756 AD failure, September 8, 2026
Open source www.sec.gov

S45 | CALM-CSU: EP262 primary controlled trial report, October 29, 2026
Open source onlinelibrary.wiley.com